Roll No. | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-6)

BIOPHARMACEUTICS AND PHARMACOKINETICS-THEORY

Subject Code : BP604T

M.Code : 77989

Date of Examination: 13-12-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION-A

1. **Explain briefly:**

   a) What is the unit of K_a?

   b) Give two examples of plasma proteins.

   c) Mention the non-renal routes of drug elimination.

   d) Mention four reasons for reduced oral bioavailability of drugs.

   e) Differentiate between one and two compartment open models with the help of representative plasma drug profiles.

   f) Mention the formula for calculating $t_{1/2}$ of drug for first order kinetics.

   g) Mention the disadvantages of oral administration of drugs.

   h) What are pharmaceutical alternatives?

   i) Differentiate between drug excretion and elimination.

   j) What is total clearance and how it is calculated?

SECTION-B

2. Explain the Michaelis-menton kinetics with the help of suitable equations. Mention the influence of Michaelis-menton kinetics on drug pharmacokinetics.

3. Explain how K_a is determined by the method of Residuals and by Wagner-Nelson method. Which of these methods gives a better estimate of K_a and why?

4. IV injection of a drug to a 70 kg patient gave drug serum concentration of 1.2 microgm/ml and 0.3 microgm / ml at 2 hr and 5 hr, respectively. Calculate $t_{1/2}$ of the drug assuming first order kinetics.

SECTION-C

5. Discuss the regulatory considerations pertaining to bioequivalence studies in India.

6. Comment on In Vitro In Vivo Correlations.

7. Discuss the role of pK_a of drug and pH of biological fluid in drug absorption.

8. Explain with the help of suitable equations the pharmacokinetics of a drug in plasma after IV administration that follows one compartment open model.

9. Comment on the role of drug-protein binding in influencing pharmacokinetics.

10. What is meant by MAT, MDT and MRT? What is the relation between them?

11. Enumerate pharmaceutical factors influencing drug absorption and discuss any two in detail.

12. Comment on factors affecting drug distribution in the body.

13. If the $t_{1/2}$ of a drug is 12 hr, how long would it take for 150 mg dose to get 30% eliminated?

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NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.

DEC-2024